CPE 420: Intro to Data Mining

19/20

HomeWork 1

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type?!

Problems

1. (4 points) For each of the following, choose **True** or **False** if the activity is a data mining task. (True if it is, False if it is not.)
   (a) Predicting the outcome of an election by analyzing the changing demographics in a region. **True**
   (b) Computing the total graduation numbers for each department at a university. **False**
   (c) Using a set of entity attributes to determine a likely classification for the entity. **True**
   (d) Using a search engine to find the address of a restaurant. **False**

2. (5 points) For each of the following, classify the attribute as **continuous or discrete** and indicate the attribute type (**nominal, ordinal, interval, or ratio**).
   (a) Splitting people based on the country they were born in. **Discrete, Nominal**
   (b) A person's weight in kilograms. **Continuous, Ratio**
   (c) Sound as measured by a loudness meter. **Continuous, Interval**
   (d) Student's GPA score. **Continuous, Interval** discrete -0.5
   (e) Length as measured by a ruler. **Continuous, Ratio**

3. (3 points) State the type of each attribute **before and after** the following transformations.
   (a) Hair color is mapped as: black = 0, brown = 1, red = 2, blonde = 3, grey = 4, white = 5.
   (b) Grade (0–100) is mapped to: A = 4.0, A- = 3.5, B = 3.0, ..., E = 0.0.
   (c) Height is changed from meters to {Short, Medium, Tall}.

a) Before: Discrete, Nominal **After:** Discrete, Nominal
b) Before: Continuous, Ratio **After:** Discrete, Ordinal
c) Before: Continuous, Ratio **After:** Discrete, Ordinal

interval -0.5

4. (8 points) This problem compares similarity and distance measures.

Let

$x = 0101010001$

$y = 0100011000$

(a) For binary data, the L1 distance corresponds to the Hamming distance; that is, the number of bits that are different between two binary vectors. The Jaccard similarity is a measure of the similarity between two binary vectors. Compute the **Hamming distance** and the **Jaccard similarity** between x and y.

Hamming Distance :

$x = 0101010001$

$y = 0100011000$

Hamming distance = 3

Jaccard Distance :
A = {2,4,6,10}
B = {2,6,7}

Union: {2,4,6,7,10}
Intersection : {2,6}

Jaccard = Intersection / Union = 2 / 5 = 0.4

(b) Which approach is more similar to the Simple Matching Coefficient, and which is more similar to the cosine measure? Explain. (Note: The Hamming measure is a distance, while the other three measures are similarities, but don't let this confuse you.)

Hamming distance is more similar to the Simple Matching Coefficient because SMC = 1 – (Hamming distance / n) so they are linearly related and both treat M00 as relevant.

While Jaccard similarity is more similar to the cosine measure because they ignore M00 and focus only on M11 (counts of 1s)

(c) Suppose that you are comparing how similar two organisms of different species are in terms of the number of genes they share. Describe which measure, Hamming or Jaccard, you think would be more appropriate for comparing the genetic makeup of two organisms. Explain. (Assume that each animal is represented as a binary vector, where each attribute is 1 if a particular gene is present in the organism and 0 otherwise.)

Jaccard similarity because it ignores M00 which in genetics would otherwise inflate similarity due to many irrelevant absent genes

(d) If you wanted to compare the genetic makeup of two organisms of the same species, e.g., two human beings, would you use the Hamming distance, the Jaccard coefficient, or a different measure of similarity or distance? Explain. (Note that two human beings share $> 99.9\%$ of the same genes.)

Hamming distance because two humans share nearly all genes so Jaccard would almost always be equal to 1 and that give us no information, While Hamming distance focuses on the rare differences, which are what matter when comparing same species.